

Advisor: PhD Ha Manh Hung

Team leader: Phan Nam Khanh - 22070980
Nguyen Duc Huy - 22070264
Nguyen Van Giang - 22070278
Trinh Van Bach - 22070946
La Nhat Bao Duy - 2071044

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MULTI-CLASS TRAFFIC VIOLATION DETECTION IN VIETNAM USING INSTANCE SEGMENTATION

BACKGROUND & PROBLEM

- Mixed Traffic Flow
- Difficult to apply in practice (legal)
- The old technology systems are now outdated.
- Lack of a closed-loop ecosystem for sanctions and complaints.

Results I: Model Performance

Metric	Object Detection (Box)	Segmentation (Mask)
Precision	0.877 (87.7%)	0.864 (86.4%)
Recall	0.806 (80.6%)	0.787 (78.7%)
F1-score	0.840 (84.0%)	0.820 (82.0%)
mAP50	0.869 (86.9%)	0.855 (85.5%)
mAP50-95	0.732 (73.2%)	0.604 (60.4%)

Results II: Violation logic & comparison

Module	TP	TN	FP	FN
Sidewalk	5	62	0	0
Red Light	2	239	0	0
Lane Violation	0	67	0	0
Wrong-way	14	1	0	0
Traffic Sign	2	18	0	0
Helmet Detection	43	685	0	1024

RESULT & EVALUATION

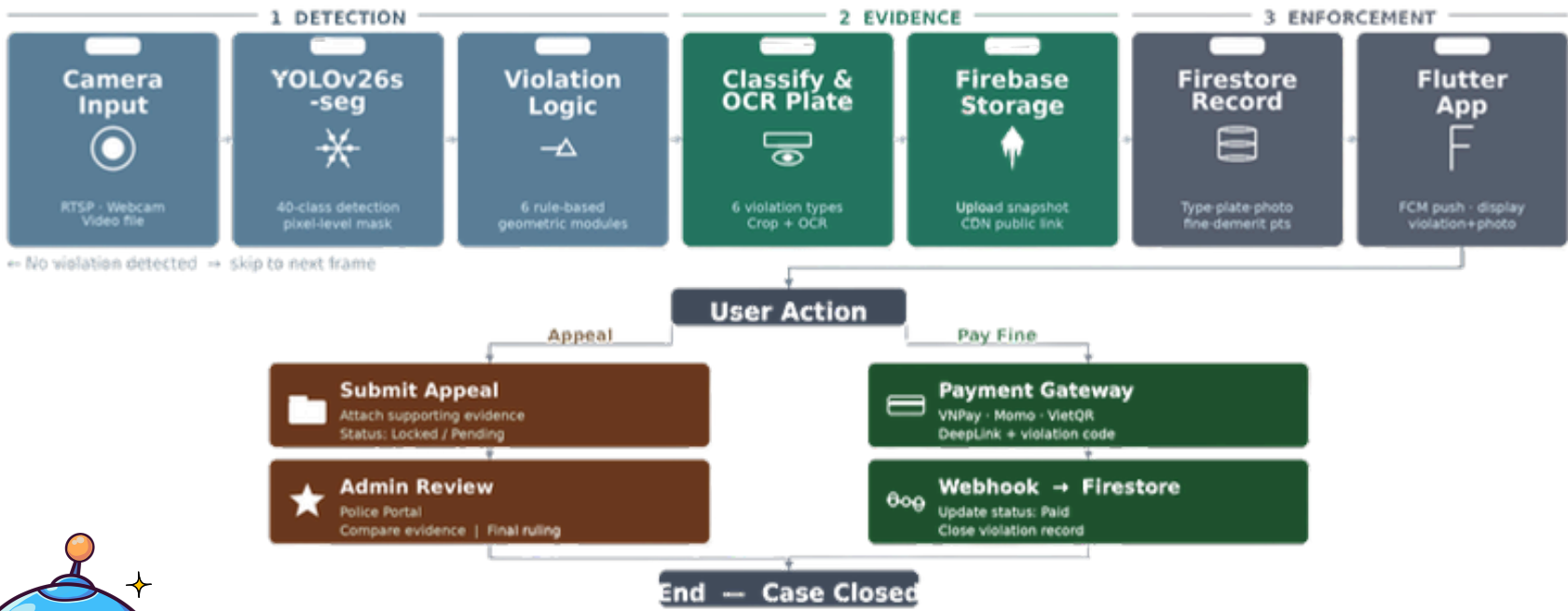
DATA

- Sources: Photos taken from pedestrian bridges, intersections, screenshots, to CCTV platforms . (IHanoi, Can Tho Smart,...)
- 40 classes - 6 core logical groups

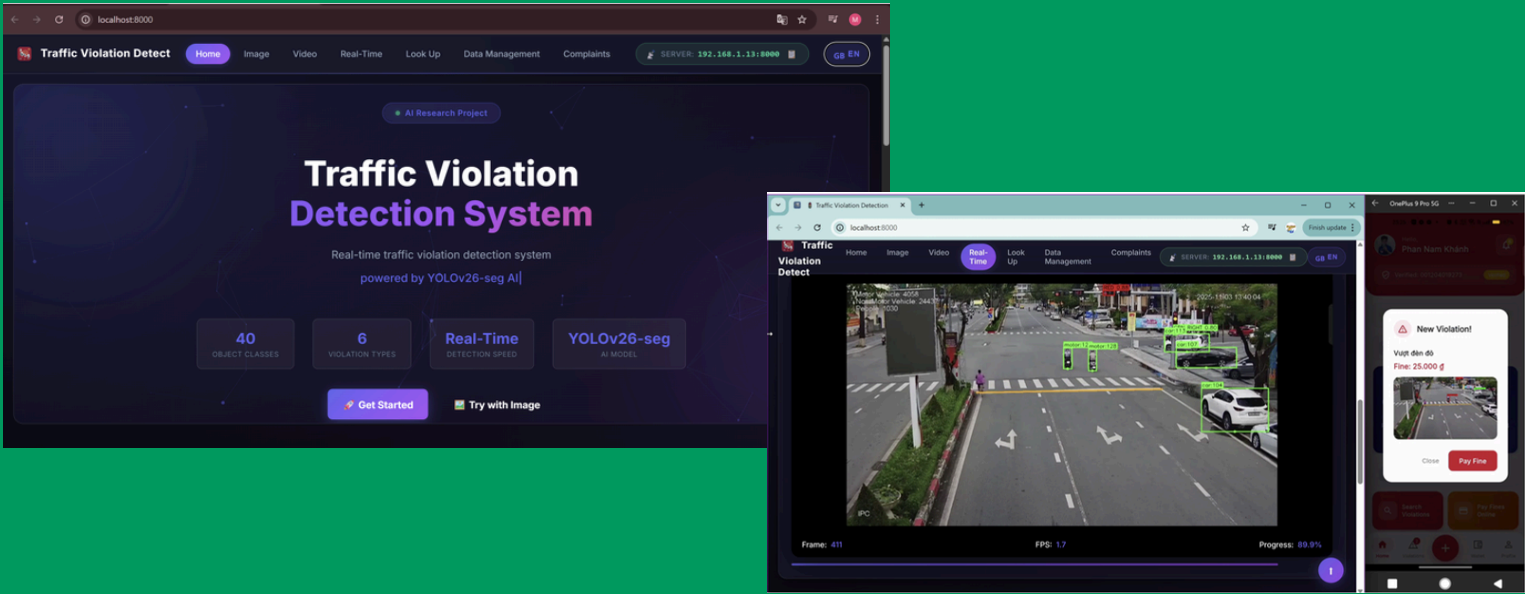
4,482
Images Captured

47,039
Stickers / Masks

Algorithm flowchart



System Implementation



NO
HELMET

RED LIGHT

SIDEWALK

LANE
MISUSE

WRONG
WAY

TRAFFIC
SIGN

CONCLUSION

- Balanced accuracy and real-time performance
- Closed-loop penalty/complaint system ecosystem
- Flexible architecture for upgrades and expansion

FUTURE WORK

- Pose Estimation Integration
- Anomaly Detection
- 3D Trajectory & Behavior Models

